

Md Kibria Saroare

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Summary

Electrical engineer and Ph.D. researcher with over eight years of experience in power generation, electrical system maintenance, troubleshooting, and engineering safety. Former senior electrical engineer at a 1,690 MW generation utility, with hands-on experience in power equipment, substations, protection, UPS, and variable-frequency drives. Current research focuses on AI data center electrical loads, power system modeling, grid disturbances, and resilient power-source transitions.

Education

Ph.D. in Engineering, Materials, Systems and Infrastructures, Louisiana Tech University In Progress
Research focus: Electrical power systems and AI data center integration. *Expected: 02/2029*

Graduate Study in Electrical Engineering, Louisiana Tech University

Bachelor's Degree in Electrical Engineering, Khulna University of Engineering & Technology Sep. 2014

Technical Skills

- **Power and Data Center Systems:** UPS, BESS, backup generation, power-source transfer, transformers, switchgear, protective relays, VFDs, electrical distribution, voltage regulation, and cooling-load modeling.
- **Engineering and Simulation Tools:** MATLAB/Simulink, Simscape Electrical, PSCAD, PSS/E, DigSILENT PowerFactory, ETAP, and Vitis HLS.
- **Engineering Analysis:** Power flow, short-circuit and fault analysis, protection coordination, dynamic load modeling, system testing, validation, and technical documentation.
- **Programming:** Python, C, C++, MATLAB, Verilog, and Lingua Franca; simulation and hardware-in-the-loop-oriented workflows.
- **Electrical Engineering Practice:** Substation equipment and layouts, cable sizing, control panels, commissioning, maintenance, root-cause troubleshooting, and safety procedures.

Professional Experience

Senior Electrical Engineer, *Ashuganj Power Station Co. Ltd.* – Bangladesh Mar. 2021 – Aug. 2025

- Supported reliability and maintenance of electrical systems at a 1,690 MW power generation utility.
- Led cross-functional troubleshooting of electrical and control equipment to restore operational continuity.

Electrical Engineer, *Ashuganj Power Station Co. Ltd.* – Bangladesh Sep. 2016 – Mar. 2021

- Modeled and analyzed electrical systems to support power plant upgrades and engineering decisions.
- Diagnosed faults in critical electrical components and supported testing, maintenance, and restoration.
- Worked with power equipment, protection, and operating procedures to maintain safe plant operation.

Lecturer, *Prime University* – Bangladesh Oct. 2014 – Sep. 2016

- Taught Power System Analysis, Electrical Machines, and Circuits; developed laboratory assessments.

Research Experience

Research Assistant, *Louisiana Tech University* – Ruston, LA Aug. 2025 – Present

- Model AI data center IT loads, UPS power conversion, and cooling systems in MATLAB/Simulink.
- Study grid disturbances, UPS ride-through, backup generation, and coordinated source transitions.
- Develop and validate power-system monitoring and disturbance-detection algorithms using simulation and FPGA tools.

Selected Projects and Publications

- **DC-CLM:** Extended WECC composite load modeling to represent AI data center UPS, cooling, and workload-dependent demand; Accepted at SmartGridComm 2026. [Code]
- **FedGuard-DC:** Developed a privacy-preserving approach to data center load forecasting and false-data-injection detection; built synthetic six-site scenarios with grid disturbances; Accepted at NAPS 2026 [Code]
- **Data Center Supervisory Control:** Coupled Lingua Franca and Simulink through UDP to study grid, UPS, and diesel-generator transitions and cooling control. [Code]
- **GridStream:** Hardware-efficient point-on-wave disturbance monitoring framework; presented at TPEC 2026.

Professional Credentials and Recognition

- Passed Fundamentals of Engineering (FE), Electrical and Computer; IEEE Power & Energy Society member.
- Awarded a \$10,000 NSF and Louisiana Board of Regents GRASS grant for a grid-monitoring proposal; TPEC 2026 NSF travel grant recipient.